Protocol for connecting ECR with POS terminal

OPERATING INSTRUCTIONS

Revision History

Revision	Description	Author	Date
2.8	INPUTDATA modified. New fileds: Minimum Input Lengths, Maximum Input Length, Mask Input Data. Input label has variable length now. New transaction types: PIN Verify Begin and PIN Verify End.	Dejan Marić	14.02.2014
2.9	Partial void functionality added. Amount sent in void request is amount to be voided.	Boris Bogdanoski	29.04.2014
2.10	Language ID added.	Dejan Marić	18.08.2014
2.11	RRN added to ECR response. Incorrect descriprtions removed.	Boris Bogdanoski	21.8.2014
2.12	Initialize (parameter download) added to supported transactions.	Boris Bogdanoski	09.09.2014
2.13	Print Data added.	Dejan Marić	23.02.2015
2.14	Print Data modified – message length increased (1000 chars), Form Feed added.	Dejan Marić	09.06.2015
2.15	1D and QR barcodes added to Print Data.	Boris Bogdanoski	01.09.2015
2.16	Cashier ID extended to 10 digits (new field). New transaction type: Current Issuer Totals.	Dejan Marić	22.02.2016
2.17	New transaction type: Current Issuer Transactions & Totals.	Dejan Marić	22.03.2016
2.18	New transaction Types: Sale+Cash and Payservices. New fields in request message. PIN Entry flag added in response message.	Dejan Marić	09.05.2016
2.19	New fields added: Transaction Amount 2 in response message for for Sale+Cash and. Payservice data for Payservices.	Dejan Marić	13.05.2016
2.20	New transaction type: Balance inquiry. New fields added in response message: Available balance, Offline cryptogram	Dejan Marić	28.06.2016
2.21	New Message Type 30 (Card data) for Card Activated Transaction Available Balance description changed	Dejan Marić	04.08.2016
2.22	Loyalty and Mobile Payment data added	Boris Bogdanoski	14.12.2016
2.23	Cancel Current Transaction added	Nemanja Marjanović	25.04.2017
2.24	Extended Hold (25) and Error (26) messages containing codes added Formatted text to print added to ECR response for printing on ECR	Boris Bogdanoski	12.07.2017
2.25	Void of Authorization added to transaction list	Boris Bogdanoski	31.08.2017
2.26	Terminal ID Transaction added	Nemanja Marjanović	15.11.2017

2.27	Sleep Transaction added	Nemanja Marjanović	22.02.2018
2.28	Bus Plus Loyalty Check added	Boris Bogdanoski	8.3.2018
2.29	Extended Hold Message list updated	Dejan Marić	2.4.2018
2.30	AS24 ECR processing and transaction type added	Nemanja Marjanović	25.06.2018
2.31	New transaction type added for AS24	Nemanja Marjanović	19.07.2018
2.32	DCC over ECR added	Nemanja Marjanović	25.07.2018
2.33	New transaction type (read payment card)	Nemanja Marjanović	18.10.2018
2.34	New transaction type (Instant Payment Inquiry)	Dejan Marić	05.12.2018
2.35	New transaction type (Restart Terminal)	Mladen Bogatinov	03.01.2019
2.36	New Card Source Data and Company Name values for Instant Payment	Dejan Marić	11.01.2019
2.37	Instant Payment Refund added, including reference number	Boris Bogdanoski	25.3.2019
2.38	New transaction type added for SOCAR	Nemanja Marjanović	29.03.2019
2.39	Increased size for reference number	Dejan Marić	08.05.2019
2.40	New field in request added: QR code for Instant Payment (PULL mode)	Dejan Marić	03.06.2019
2.41	New transaction types added for reading Personal Card and Vehicle Card for Serbian documents	Nenad Pavlović	23.09.2019.
2.42	Description of physical interface updated	Dejan Marić	05.12.2019
2.43	New transaction type (Check PIN for Lukoil)	Dejan Marić	20.12.2019
2.44	Mifare card handling added	Boris Bogdanoski	30.01.2020
2.45	Terminal functions processing added	Nemanja Marjanović	05.03.2020
2.46	Specific processing flag added	Nemanja Marjanović	14.04.2020
2.47	Radcom / transportation functionality	Boris Bogdanoski	01.06.2020
2.48	Read payment card specific processing added	Nemanja Marjanović	12.06.2020
2.49	F4G for Instant payment added	Dejan Marić	14.10.2020
2.50	Radcom / transportation functionality (renumbered as 2.50, originally added as 2.49)	Boris Bogdanoski	12.10.2020
2.51	Radcom – new tag added (authenticity token)	Nemanja Marjanović	24.11.2020
2.52	Size of DCC Number updated	Dejan Marić	05.03.2021

2.53	Acquirer ID added to response	Boris Bogdanoski	10.3.2021
2.54	ECR print data – new QR format added	Dejan Marić	18.10.2021
2.55	Transaction ID added	Dejan Marić	27.12.2021
2.56	Update codes for Extended Error messages	Dejan Marić	13.05.2022
2.57	New transaction: Refund Specific added	Dejan Marić	30.08.2022
2.58	Radcom – new tag added (Hashed PAN)	Nemanja Marjanović	13.09.2022
2.59	Transaction ID size increased New transactions (E-Money Inquiry and E-Money Refund)	Dejan Marić	27.10.2022
2.60	Read Mifare UID Use RRN for Refund and Preauthorization Completion	Dejan.Marić	09.12.2022
2.61	DCC disclaimer text increased. New fields: SPDH Sequnce number, Installment Delay and Creditor type added	Dejan Marić	27.01.2023
2.61a	Field separator after Installments Delay added	Dejan Marić	23.02.2023
2.61b	Added explanations: Last ACK not received from cash register, terminal doesn't send void to host. Transaction is approved when value of Transaction flag is 01 or 02.	Zvonimir Kovačić	09.01.2024.
2.62	Tip processing for restaurants supported	Dejan Marić	18.03.2024

1. PURPOSE

Purpose of this document is supporting ECR developers while developing ECR connection with POS terminal.

2. DESCRIPTION OF PROCESS

2.1. Introduction

This document describes the messages that are exchanged between an external application and a POS terminal.

Tipically POS transaction scenario is following :

A part of the transaction - typically the transaction type and the amount - is inputted on the external interface. Data not given by external application - but needed for POS transaction - is inputted on the POS terminal. Tipically card number is keyed into POS terminal or magnetic card is drawn through magnetic card reader or chip card is inserted into chip card reader.

The transaction is initiated by the external interface. POS terminal requests card number. The POS terminal is requesting POS authorization after receiving card number in one of previous described ways. POS terminal answers to external application after receiving authorization response. External application receives data with transaction response and receipt data including the authorization code if any.

In some specific configuration transaction can start from terminal side. It is card activated transaction, optionaly used for unattended terminals. Terminal is in waiting state (waits for card swipe, insert, present or request data from external application). Once card is read terminal sends card data to external application. External application can verify card data and send transaction request (with amount, transaction type, ETC) to terminal. Further transaction flow is as same as described in previous section.

2.2. Physical Link

2.2.1. Communication Interfaces

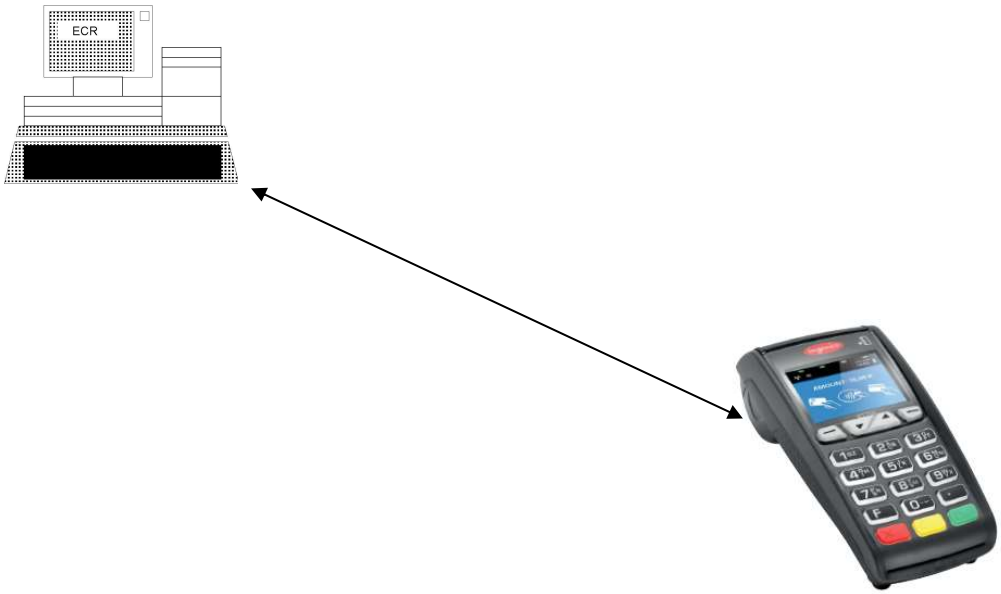
The Physical Link to the POS terminal will be implemented by one of following communication interfaces:

- RS-232
- USB
- Ethernet/Wifi

Data transmission to and from the terminal is Asynchronous regardless of communication interface.

If RS-232 interface is used configuration (configured speed and data format) must be the same at both sides:

Speed : *1200 -115200 Bps*
Data Format : *7 bits , parity even , 1 stop bit or*
: *8 bits, parity none, 1 stop bit*
Flow Control : *None*



2.2.2. Message block format

STX	MESSAGE BLOCK	ETX	LRC
1	variable, 256 maximum	1	1

2.2.3. Communications Characters Used

Character	Hex Value	Meaning
STX	02	Start of text
ETX	03	End of Text
EOT	04	End Of Transmission
ACK	06	positive ACKnowledgment
NAK	15	Negative AcKnowledgment

2.2.4. LRC - Longitudinal Redundancy Check

The LRC character provides an error detection facility.

The LRC is generated by Exclusive-OR'ing all data bytes following the STX, up to and including the ETX. LRC character check as per ISO 1155. LRC is an XOR calculated on:

<Message DATA><ETX>

The LRC byte follows the ETX, and must have the same parity as the rest of the message.

2.2.5. Message Format

The format of the actual data sent by the ECR:

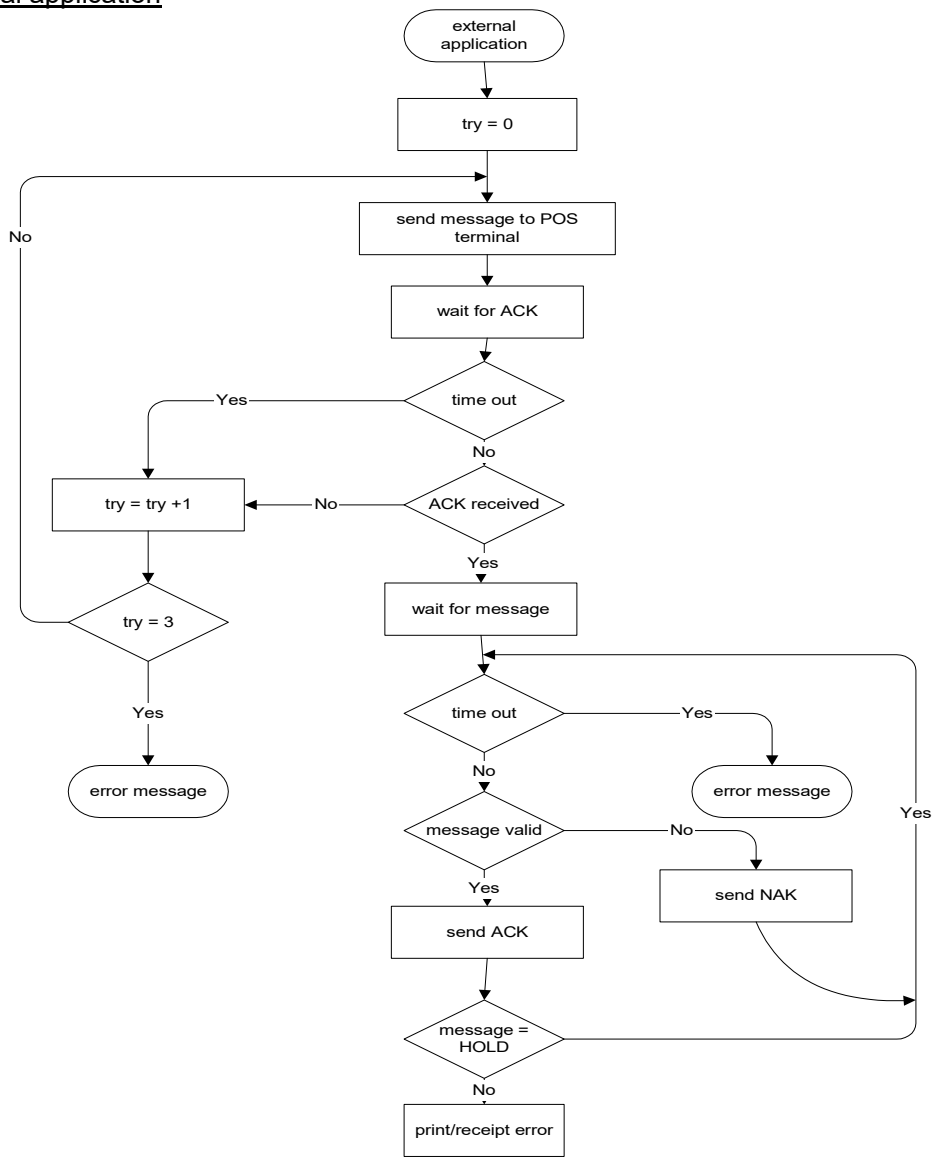
<STX><STX><Message DATA><ETX><LRC>

The format of the actual data received by the ECR:

<STX><Message DATA><ETX><LRC>

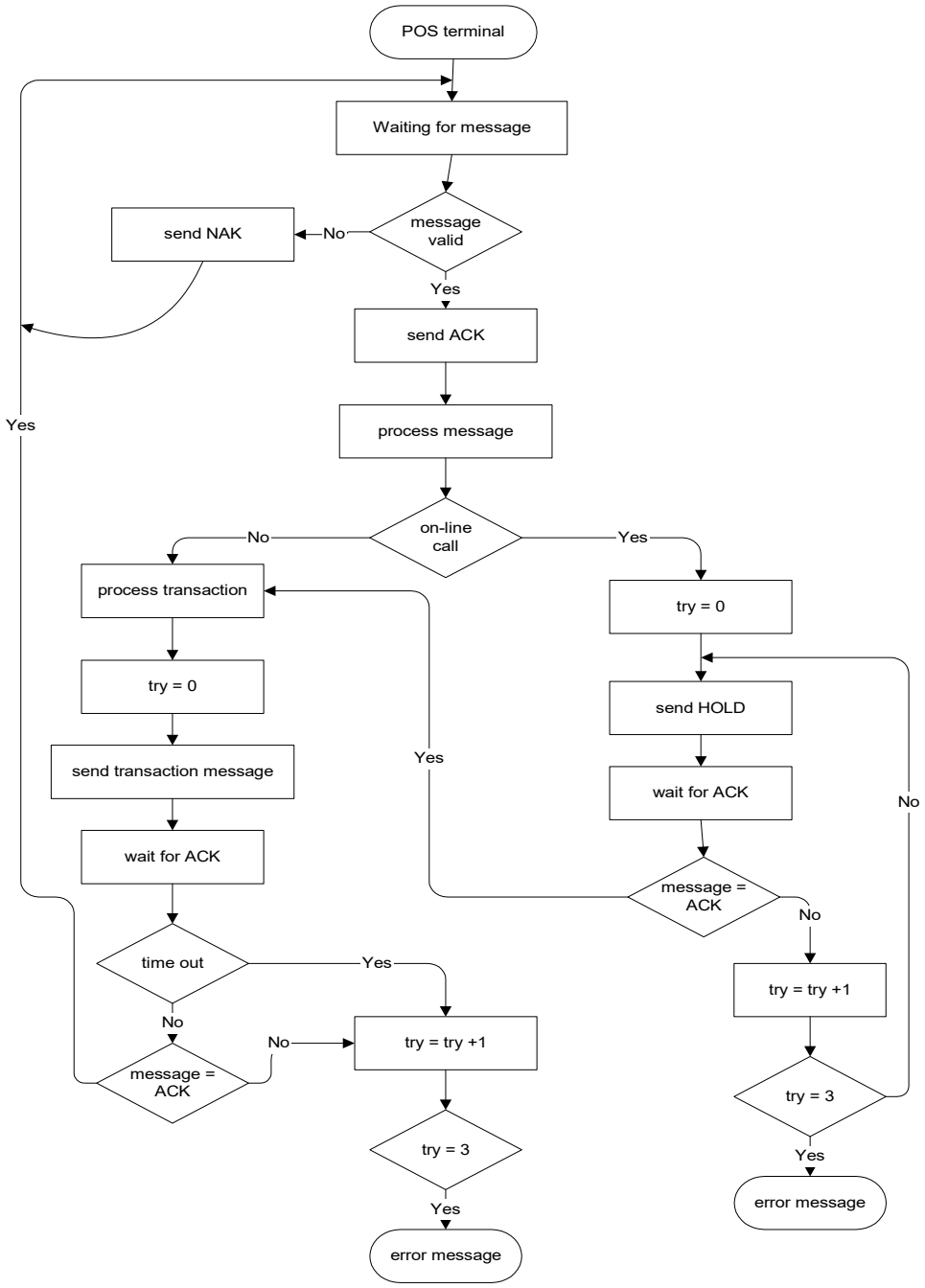
2.2.6. Communication sequence

2.2.6.1. External application



Picture : External application communication sequence

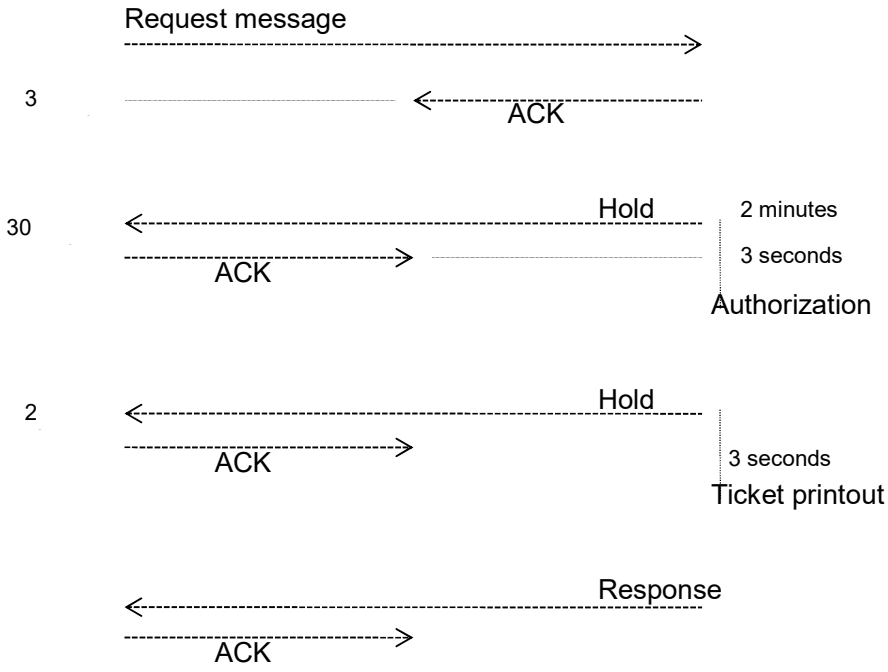
2.2.6.2. POS terminal



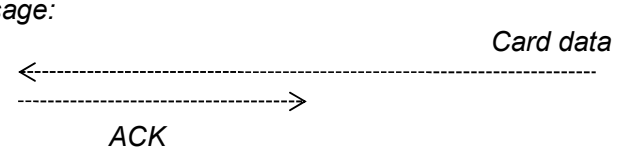
Picture : POS terminal communication sequence

2.2.7. Example

Cash Register Terminal



If transaction is activated by the terminal card data is provided to Cash register prior to request message:



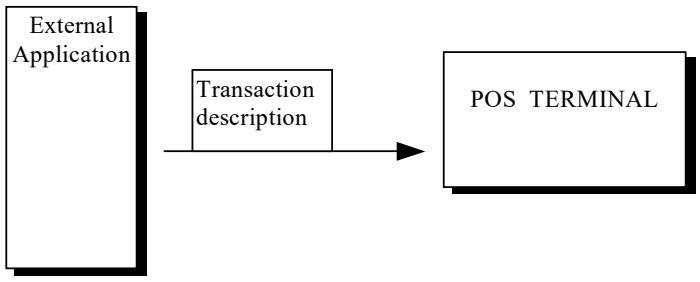
- Minimum ACKnowledgment time = 0ms.
- Maximum ACKnowledgment time = 1000ms.
- If no acknowledgment is received before the time expires then it is treated as a NAK and the message is retransmitted.
- Maximum number of retries = 3
- If POS terminal doesn't receive last acknowledgment from the cash register, after 3 retries, transaction will be approved on the POS side and terminal WILL NOT send reversal / void to host. In that case transaction will stay APPROVED.
- Reversal is only generated in case that POS terminal didn't receive response from host that transaction has been processed.

2.2.8. The external transaction protocol overview

2.2.8.1. Description

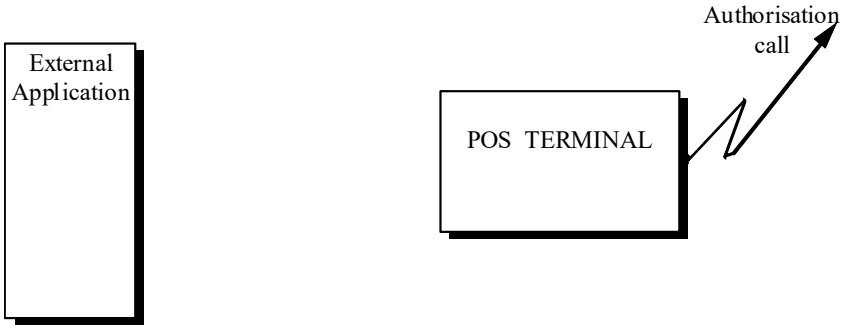
Transaction :

The external application generates transaction message to the POS terminal. The POS terminal will acknowledge the request prior to processing the transaction.

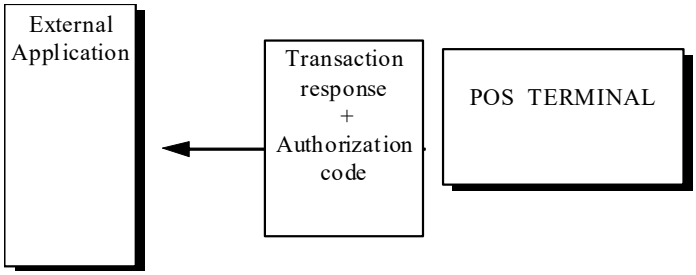


The POS terminal receives this transaction and initiates the transaction process. The transaction amount is included in this message and all other data required will be prompted on the POS terminal. The POS terminal examines the content of the transaction request, asks for not given values (Ex : Card Number) and performs the authorization operations (e.g. asks for PIN code, performs authorization call, checks stoplist...).

In accordance with international card standards : reading card data on magnetic card readers or chip readers other then those on POS terminal is in contradiction with international card standards.



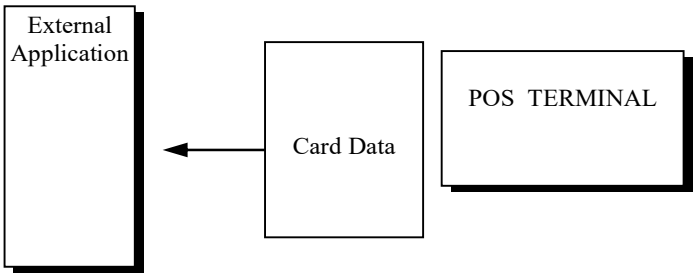
The POS terminal sends the transaction description back to the external application with an transaction response including authorization code or deny message.



Important Remarks:

- Every message from the external application contains a sequence number that is incremented every time a transaction is successfully done with the POS terminal. If the POS terminal receives a request with the same sequence number as the previous one, it knows that transaction has been already treated and responds with the same data as previous request..
- When the POS terminal have to make external authorization call or data entry, hold messages may be sent to the external application to inform it to wait longer for response.

For Card Activated transaction, terminal can send card data to external application. External application is in waiting state, here.



Once card data is received external application can generate transaction message. In this configuration, transaction is initiated by the terminal. Terminal is establishing communication link over serial interface. If Ethernet interface is used, communication link is established by external application since terminal works in ethernet server mode. It could be done once (for example: after restart of both devices).

Transaction types :

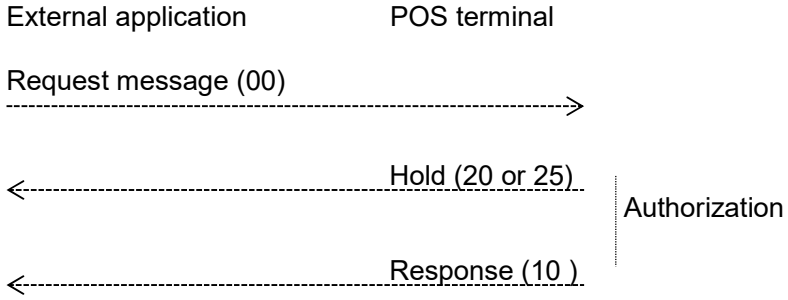
Transactions are split in two categories :

Type 1 : New transaction entries (SALE , ...)

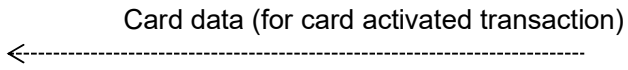
Type 2 : Action on previously stored transactions (VOID,...)

2.2.9. Examples

2.2.9.1. Transaction with Authorization



In specific configuration there could be additional message:



2.2.10. Message format

All fields use ASCII character set.

2.2.10.1. Request messages to the POS terminal

Message 00: Request for a transaction

	Name	Condition	Size	Format	Default Value	Value
1.	**Identifier**	Mandatory	2	Numeric	00	Always **00**
2.	**Terminal ID**	Mandatory	2	Numeric	00	ID of the terminal (default value **00**) If the value is different than 00, it indicates the Acquirer ID in case of multiacquirer configuration. It may be used for different insurance types.
3.	**Source ID**	Mandatory	2	Numeric	00	ID of the source (default value **00**)
4.	**Sequential number**	Mandatory	4	Numeric		Number between 0000 and 9999 incremented for each message sent is handled by external application POS terminal will check if the sequential number already exists and it will return the same data in such case. If sequential number is 0000, it will be ignored. Sequential number is not checked for Settlemernt and Batch Status transactions.

	Name	Condition	Size	Format	Default Value	Value
5.	**Transaction type**	Mandatory	2	Numeric	01	**00** : Initialization (parameter download) **01** : Sale **02***:* Preauthorization **03:** Cash Advance **04:** Preauthorization Completion **05:** Refund **06:** Offline Sale **07:** Sale+Cash **08:** Payservices **09:** Balance Inquiry **10:** Void transaction **11:** Cancel previous transaction **12:** Void of Authorization/Preauthorization **20:** Sale with Installments **21:** Refund Specific (ING) **30:** Settlement **31:** Batch Status **32:** Current Totals **33:** Current Issuer Totals **34:** Current Issuer Transactions & Totals **77:** Read Mifare UID **78:** Read Personal card data **79:** Read Vehicle card data **80:** Pin Verify Begin **81:** Pin Verify End **82:** Restart Terminal **83:** Check PIN (Lukoil) **84:** E-Money Refund **85:** E-Money Inquiry **88:** Socar Transaction **89:** Instant Payment Refund **90:** Instant Payment Inquiry **91:** Read payment card **92:** AS24 Fleet Card Transaction Completion **93:** AS24 Fleet Card Transaction **94:** Bus Plus Loyalty Check **95:** Sleep transaction **96:** Get Terminal ID **97:** Print Data (transaction) **98:** Card Read **99:** Input Data
6.	**Printer flag**	Mandatory	1	Numeric	1	**0:** Receipt printed by the ECR **1** : Receipt printed by the POS terminal
7.	**Cashier ID**	Mandatory	2	Numeric	00	ID of the cashier (default value **00**)
8.	**Transaction number**	Optional	0 or 6	Numeric		Empty for type 1 transactions Needed for Void
9.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	

	Name	Condition	Size	Format	Default Value	Value
10.	**Transaction amount 1**	Optional	12 max	Numeric		Amount of the transaction. In case of void, it should contain amount to be voided. Note: This field is not used for Initialization, Batch Status, Settlement, Current Totals
11.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
12.	**Field separator**	Mandatory	1	FS (chr28)		
13.	**Amount exponent**	Required if the field **Transaction amount 1** is filled	0 or 2	Sign + Numeric	+0	Default value is : +0. Note: This field is not used for Initialization, Batch Status, Settlement, Current Totals
14.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
15.	**Amount currency**	Required if the field **Transaction amount 1** is filled	0 or 3	Numeric		ISO Currency code for multi currency applications. Note: This field is not used for Initialization, Batch Status, Settlement, Current Totals
16.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
17.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
18.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
19.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
20.	**Authorization code**	Optional	0-6	Alpha Numeric		Authorization code for Offline Sale and Refund Specific
21.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
22.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
23.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
24.	**Input label**	Optional	0-16	Alpha Numeric		Label for Input Data
25.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
26.	**Insurance policy number**	Optional	0-20	Alpha Numeric		Insurance policy number which may be transferred to the host and printed on the receipt
27.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
28.	**Installments number**	Optional	0-2	Alpha Numeric		Number of installments

	Name	Condition	Size	Format	Default Value	Value
29.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
30.	**Minimum Input length**	Optional	2	Numeric	01 (04)	Minimum Length of Input Data
31.	**Maximum Input length**	Optional	2	Numeric	20 (12)	Maximum Length of Input Data
32.	**Mask Input Data**	Optional	1	Numeric	0	0- Input data not masked 1- Input data masked
33.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
34.	**Language ID**	Optional	2	Numeric	00	00=Default Terminal Language 01=Macedonian, 02=Serbian, 03=English, 04=Romanian, 05=Montengro, 06=Bosnia&Herzegovina 07=Croatian 08=Slovene, 09=Albanian, 10=Bulgarian, 11=French, 12=Slovak, 13=German, 14=Italian, 15=Spanish, 16=Malagasy...
35.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
36.	**Print Data**	Optional	1000	Alpha Numeric + CR, FF		Data that should be printed on receipt. Data could be delimited by Carriage Return or Form Feed
37.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
38.	**Cashier ID 2**	Optional	0-10	Numeric		ID of the cashier (extended)
39.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
40.	**Transaction amount 2**	Optional	12 max	Numeric		Additional amount of the transaction. In case of Sale+Cash transaction this field contains Cash amount. In case of Sale + Tip this field contains Tip amount.
41.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
42.	**Payservices Data**	Optional	600 max	Alpha-numeric		TLV based data (Product code, service value,...)
43.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
44.	**Transaction Activation Code**	Optional	16 max	Numeric		Transaction Activation Code for Mobile Payment. Regular purchase request shall be sent with TAC value to perform Mobile Payment
45.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
46.	**Instant Payment Reference**	Optional	35 max	Numeric		Reference number for Instant Payment – mandatory for Instant Payment Refund (same value as in response on Instant Payment Purchase or Inquiry)

	Name	Condition	Size	Format	Default Value	Value
47.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
48.	**QR code data**	Optional	200 max	Alpha-numeric + spec chars		QR code data for Instant Payment (Pull mode).
49.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
50.	**Specific processing flag**	Optional	2	Numeric		01 : SBER request 02 : Radcom / transportation deferred authorisation 03 : Special read card request 04 : Remove card request 05 : Radcom / transportation ticket validation – card read
51.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
52.	**Radcom / transportation specific data**	Optional	1000 max	Alpha-numeric		Specific Radcom / transportation related data
53.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
54.	**Transaction ID**	Optional	30 max	Alpha-numeric		Transaction identifier (for specific implementation)
55.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
56.	**RRN**	Optional	12	Numeric		Retrieval Reference Number (for Refund, Preauthorization Completion and Refund Specific transaction)
57.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
58.	**Credit type**	Optional	1	Alpha-numeric		Type of creditor for Installments (possbile values **S** – Store **F** – Financial institution)
59.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
60.	**Installments Delay**	Optional	3	Alpha-numeric		Postpone for Installments. Format is: Dxx or Mxx where first character represents postpone type (**D**ays or **M**onths) and xx is postponement (two digit number)
61.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	

Notes

- If communication between ECR and the terminal is broken during the transaction and final response is not received by the ECR, the status can be checked from the ECR by sending the same request with the same amount and sequential number and the terminal will respond with the same data if the transaction exists. If the transaction

with the same sequential number does not exist on the terminal, it will be considered as new transaction request and processed in common way

- Sequential number is not processed in case of Settlement and Batch Status transactions of if 0000 is sent as sequential number.
- If the status response is not received by the ECR and the customer decides to either cancel the transaction or pay with cash, the ECR should send the Cancel previous transaction request with the same seqential number before next transaction request until it is resolved. The terminal will reverse the tranaction with that sequential number if it exists. The positive response will be returned by the terminal in any case.
- Settlement request will initiate settlement of all acquirers on the terminal.
- Batch Status request is sent from the ECR to check the status of the last settled batches for all the acquirers.
- Current Totals request is used to obtain either overall current totals on the terminal if 00 is sent for cashier number or current totals by cashier if cashier number is specified in the request.
- Card Read request is used to read full card number from the specific local cards with card number length up to 8 digits.
- PIN Verify performs offline pin checking and informs ECR if it is sucessfully verified by chip card or not.
- Max and Min length of Input Data could be defined by ECR. Minimal length is not checked at external pinpad. Minimum length for external pinpad is always 4. Maximal length at external pinpad could be up to 12 but for terminal it could be up to 20.
- Input data could be masked or visible on teminal. Input Data is always masked on external pinpad.
- In case of Void, if entire transaction is to be voided, amount field must contain the same value as original transaction. If only partial void is requested, the amount sent in void request should contain the amount to be voided and, practically, the transaction will be adjusted to new amount which is equal to difference between original and void amount.
- Print Data – Terminal will print received data from ECR. Data will be printed at the end of slip if provided for common payment transactions. If Print Data transaction is used terminal will print this data only (without any transaction processing ie. no amount entry, no communication with host,...). Regardless of transaction type data shuold be in following format:
  <font type><data><CR<font type><data><CR>...<font type><data><*FF*>

  - <font type> : '1' – Normal (40 characters Hypercom, 48 characters Ingenico)
    '2' – Bold (20 characters Hyperom, 24 characters Ingenico)
    '3' – Small (48 characters at Ingenico, only)
    '4' – 1D barcode (Ingenico only)
    '5' – QR code (Ingenico only)
    '6' – QR code2 (Ingenico only) – QR Code with header
  - <data> : ASCII characters. It has following meaning:
    - <data> = <information data> - for font types 1, 2, 3, 4 and 5.
    - <data> = <header><information data> - for font type 6.

    - <information data> - ASCII characters. Up to 48 characters for font types 1, 2 and 3. Up to 14 characters for font type 4. Up to 900 characters for font type 5 and up to 895 characters for font type 6.
    - <header> = WWWMC – 5 ASCII numbers with meaning:

WWW – 3 ASCII numbers. It represents QR code maximum width (in pixels). Maximum value is '384' pixels.

M – 1 ASCII number which represents QR encoding mode. It has following values:

'0' – numeric
'1' – alphanumeric
'2' – byte

C - 1 ASCII number that represents QR error correction level. It has following values:

'0' – level L (lowest, 7 % of recovery)
'1' – level M (15 % of recovery)
'2' – level Q (25 % of recovery)
'3' – level H (highest, 30 % of recovery)

- <CR> : Carriage Return (0x0D) –separator between lines
- *<FF>* : Form Feed (0x0C) – Optional data. If present terminal will print form feed (move paper up for cutting). This character is mandatory in last message sent from ECR (message with Print Data).
  Example: If 1500 characters must be printed then first message will contain up to 1000 characters without Form Feed. Rest of characters will be sent in second message with Form Feed at the end.

- Cashier ID 2 - Specific requirement to extend cashier ID. Because of backward compatibility new field is added. Therefore if cashier ID is between 99 and 9999999999 it should be sent in this field. Cashier ID field should keep default (00) value, then.
- Payservices Data in request message are TLV based. TLV format is similar to EMV data (description of TLV format is above). Tags and fields are described in Appendix A of this document.
- Cancel Current Transaction can be only used until card is not read. If extended hold messages are used, cancel can be sent before hold message with code 54 – ECR_HOLD_CARDINSERTED is received.
- If preauthorzations are stored in the batch, preauthorization completion and void of preauthorization requests shall contain Transaction/invoice number. All other information will be handled by the POS terminal without card read.
- Check PIN performs online pin checking and informs ECR if it is sucessfully verified by Host or not. (Lukoil specific function).
- Radcom / transportation specific data is TLV (similar to EMV data) composite data field. It is used to transfer data related to transportation transactions in both request, response and other specific messages. Definition of this field is given in Appendix E.

Messages from the POS terminal

Message 10: Answer to a transaction message

This message is send back from the POS terminal as consequence of a request message (type 00). It contains the complete description of the transaction result with the optional authorization code and all necessary field to build the ECR receipt.

	Name	Condition	Size	Format	Default Value	Value
1.	**Identifier**	Mandatory	2	Numeric	10	Always **10**
2.	**Terminal ID**	Mandatory	2	Numeric		Value from the request message
3.	**Source ID**	Mandatory	2	Numeric		Value from the request message
4.	**Sequential number**	Mandatory	4	Numeric		Value from the request message
5.	**Transaction type**	Mandatory	2	Numeric		Value from the request message
6.	**Transaction flag**	Mandatory	2	Numeric	02	**00** : Error in the format of the request message **01** : Transaction accepted without authorization code **02** : Transaction accepted with authorization **04** : Transaction refused **05**: Terminal ID **06** : Communicaation error **07**: Sleep mode **08**: Personal/Vehicle card data **09:** Mifare card approached to reader
7.	**Transaction number**	Mandatory	6	Numeric		Transaction number assigned by the terminal
8.	**Batch number**	Mandatory	4	Numeric		Internal batch number of the terminal
9.	**Transaction date**	Mandatory	6	DDMMYY		Transaction date
10.	**Transaction time**	Mandatory	6	HHMMSS		Transaction time
11.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
12.	**Transaction amount 1**	Mandatory	12 max	Numeric		Amount of the transaction.
13.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
14.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
15.	**Amount exponent**	Required if field amount1 is not empty	2	Sign + Numeric	+0	Default value is : +0

	Name	Condition	Size	Format	Default Value	Value
16.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
17.	**Amount currency**	Required if field amount1 is not empty	3		705	Code ISO of the currency of the transaction amounts 1.
18.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
19.	**Card Data source**	Optional	0 or 1	Alpha-numeric		**M** : Magnetic stripe **S** : Smart Card **K:** Manual PAN entry **C:** Contactless **I:** Instant Payment E: E-Money
20.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
21.	**Card Number**	Mandatory	19 max	Numeric		Number of the customer's card (masked), Customer Account for Instant Payment
22.	**Field separator**	Mandatory	1	FS(chr28)	FS (chr28)	
23.	**Expiration date**	Mandatory	4	MMYY		Expiration date of the customer's card (masked)
24.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
25.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
26.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
27.	**Authorization code**	Required if field 6=01	6 max	Alpha-numeric		Authorization code (if any)
28.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
29.	**TID number**	Mandatory	8 max	Numeric		Terminal identification number
30.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
31.	**MID number**	Optional	15 max	Alpha-numeric		Merchant identification number
32.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
33.	**Company name**	Optional	16 max	Alpha-numeric		Name associated with the card type i.e.: MASTERCARD MAESTRO VISA DINERS AMEX Acquirer name for Instant Payment
34.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	

	Name	Condition	Size	Format	Default Value	Value
35.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
36.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
37.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
38.	**Field separator**	Mandatory	1	FS(chr28)	FS (chr28)	
39.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
40.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
41.	**Display Message**	Optional	25 max	Alpha-numeric		Message that will be displayed
42.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
43.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
44.	**Input Data**	Optional	20 max	Alpha-numeric		Additional Input Data
45.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
46.	**EMV Data**	Optional	500 max	Alpha-numeric		TLV based EMV data, according to EMV specificatons, required for receipt printing i.e.: Cryptogram type, cryptogram, AID, application label...
47.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
48.	**Signature line print flag**	Optional	1			0: Signature required 1: Signature not required
49.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
50.	**Acquirer Name**	Optional	10	Alpha-numeric		Acquirer name mandatory for Settlement and Batch Status requests
51.	**Debit Transaction Count**	Optional	4	Numeric		Debit transaction count mandatory for Settlement and Batch Status requests (including Sale, Cash, Sale with installments...). Voided transactions are not included.
52.	**Debit Transaction Amount**	Optional	12	Numeric		Debit transaction total mandatory for Settlement and Batch Status requests (including Sale, Cash, Sale with installments...). Voided transactions are not included.
53.	**Refund Transaction Count**	Optional	4	Numeric		Refund transaction count mandatory for Settlement and Batch Status requests. Voided transactions are not included.

	Name	Condition	Size	Format	Default Value	Value
54.	**Refund Transaction Amount**	Optional	12	Numeric		Refund transaction total mandatory for Settlement and Batch Status requests. Voided transactions are not included.
55.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
56.	**More messages flag**	Optional	1			**0:** No more messages **1:** More messages to follow (for Settlement, Batch Status requests, Current Totals, ...)
57.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
58.	**Installments number**	Optional	0-2	Alpha Numeric		Number of installments
59.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
60.	**Full Response Code**	Optional	6 max	Alpha Numeric		Response code internal and from the host for printing out
61.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
62.	**Transaction status**	Optional	2	Alpha Numeric		Transaction status information for printing out including online, offline, manual entry etc.
63.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
64.	**SPDH Terminal Totals**	Optional	69	Alpha Numeric		SPDH Terminal Totals in case of totals mismatch
65.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
66.	**SPDH Host Totals**	Optional	69	Alpha Numeric		SPDH Host Totals in case of totals mismatch
67.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
68.	**PIN Block**	Optional	16	Numeric		PIN Block for local cards to be verified on ECR including SLE4442 memory cards
69.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
70.	**Cardholder Name**	Optional	26 max			Cardholder name for printout
71.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
72.	**RRN**	Optional	12	Alpha Numeric		Retrieval Referrence Number (ISO8583 host protocol)
73.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
74.	**PIN flag**	Optional	1			0: PIN not entered 1: PIN entered
75.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	

	Name	Condition	Size	Format	Default Value	Value
76.	**Transaction amount 2**	Optional	12 max	Numeric		Additional amount of the transaction. In case of Sale+Cash transaction this field contains Cash amount. In case of Sale + Tip this field contains Tip amount.
77.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
78.	**Payservices Data**	Optional	600 max	Alpha-numeric		TLV based data (Product code, service provider information,...)
79.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
80.	**Available balance**	Optional	41 max	Alpha-numeric		Balance of Cardholder account
81.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
82.	**Offline cryptogram**	Optional	32	Numeric		Used for Borica host protocol, only
83.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
84.	**Loyalty Data**	Optional	1000 max	Alpha-numeric		TLV based data (loyalty information)
85.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
86.	**Formatted text to print**	Optional	800 max	Alpha-numeric		Formated data to print on ECR
87.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
88.	**DCC Number**	Optional	16	Numeric		DCC Number
89.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
90.	**DCC Provider**	Optional	30 max	Alpha-numeric		DCC Provider
91.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
92.	**DCC Exchange Rate**	Optional	23 max	Alpha-numeric		DCC exchange rate
93.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
94.	**DCC Exchange Rate Date**	Optional	10	Alpha-numeric		DCC exchange rate date
95.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
96.	**DCC Mark Up Percent**	Optional	8 max	Alpha-numeric		DCC mark up percent
97.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
98.	**DCC Amount**	Optional	12 max	Numeric		DCC Amount
99.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	

	Name	Condition	Size	Format	Default Value	Value
100.	**DCC Disclaimer**	Optional	1000 max	Alpha-numeric		DCC Disclaimer
101.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
102.	**DCC Status**	Optional	2	Alpha-numeric		DCC status ('A ' – Accepted, 'D ' –Declined)
103.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
104.	**DCC Currency Symbol**	Optional	3	Alpha-numeric		DCC currency symbol
105.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
106.	**Instant Payment Reference**	Optional	35 max	Numeric		Reference number for Instant Payment – generated by terminal on Instant Payment Purchase or Inquiry. The same value is required in Instant Payment Refund request
107.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
108.	**Personal/Vehicle card data**	Optional	5000 max	Aplha-numeric		TLV based data (after initiated transaction type 78 and 79).
109.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
110.	**Mifare card type**	Optional	2	Aplha-numeric		Type of the Mifare card: MF_CLASSIC 0x40 MF_ULTRALIGHT 0x10 MF_4K 0x08 MF_MINI 0x04 MF_PLUS_L2 0x02
111.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
112.	**Mifare card UID**	Optonal	32 max	Aplha-numeric		UID of Mifare card
113.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
114.	**F4G code**	Optional	35 max	Aplha-numeric		F4G code for specific Instant Payment implementation
115.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
116.	**Radcom / transportation specific data**	Optional	1000 max	Aplha-numeric		Specific Radcom / transportation related data
117.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
118.	**Acquirer ID**	Optional	2	Numeric		Acquirer identifier – present in all transaction responses and transaction reports. Might be used to determine which acquirer/merchant is used for transactions and for matching issuer and acquirer totals

	Name	Condition	Size	Format	Default Value	Value
119.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
120.	**Transaction ID**	Optional	36 max	Alpha-numeric		Transaction identifier (for specific implementation)
121.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
122.	**Credit type**	Optional	1	Alpha-numeric		Type of creditor for Installments (possbile values: **S** – Store **F** – Financial institution)
123.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
124.	**Installments Delay**	Optional	3	Alpha-numeric		Postpone for Installments. Format is: Dxx or Mxx where first character represents postpone type (**D**ays or **M**onths) and xx is postponement (two digit number)
125.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	
126.	**SPDH sequence number**	Optional	9	Numeric		Sequence number for SPDH host protocol
127.	**Field separator**	Mandatory	1	FS (chr 28)	FS (chr28)	

Notes:

- EMV Data included in the response are in EMV specific TLV format. The transferred tags may vary. Usually, it shall contain the data required for the receipt printing such as: 9F26 Application Cryptogram, 9F27 Cryptogram Information Data, 84 Dedicated File Name, 9F12 Application Preferred Name. Besides that, some other useful tags may be included: 95 Terminal Verification Results, 9F34 Cardholder Verification Method Results, 9B Transaction Status Information.
- TLV format of EMV Data field is as following:
  - TLV **T**ag, **L**ength, **V**alue
  - **T** - The tag field T consists of one or more consecutive bytes. It encodes a class, a type and a number.
  - **L** - The length field consists of one or more consecutive bytes. It encodes an integer L. In short form, the length field consists of a single byte where the bit B8 shall be set to 0 and the bits B7-B1 shall encode an integer equal to the number of bytes in the value field. Any length from 0-127 can thus be encoded by 1 byte. In long form, the length field consists of a leading byte where the bit B8 shall be set to 1 and the B7-B1 shall not be all equal, thus encoding a positive integer equal to the number of subsequent bytes in the length field. Those subsequent bytes shall encode an integer equal to the number of bytes in the value field. (see ISO/IEC 8825).
  - **V** - If L is not null, then the value field V consists of L consecutive bytes. If L is null, then the data object is empty: there is no value field.
- In case of Settlement request the terminal will initiate settlement for all acquirers.
- In case of Settlement and Batch Status requests acquirer name, date, time, batch number and totals of the last successful settlement for the single acquirer will be sent in one response message and more messages flag will be set to 1. After all acquirer

response messages, overall total message with more messages flag set to 0 will be sent. For the overall totals message string 'TOTAL ' will be used as acquirer name, all zeroes will be sent for batch number.

- In case of Current Totals request the terminal will return current number of transactions and totals by cashier if cashier number is sent from the ECR or overall totals if 00 is sent as cashier ID. 'TOTAL ' will be used as acquirer name and all zeroes will be returned for batch number.
- For Settlement and Batch Status messages if SPDH host protocol is used and totals on the terminal and host sides are not matched, terminal will provide both terminal and host totals according to SPDH format. These fields include counts and amounts of debits, credits, and adjustments. These totals contain a sign character (+ or –) in the first byte of each amount field. The format is as follows:
    - Number of debits in batch (4)
    - Amount of debits in batch (19)
    - Number of credits in batch (4)
    - Amount of credits in batch (19)
    - Number of adjustments in batch (4)
    - Amount of adjustments in batch (19)
- In case of Current Issuer Totals request the terminal will return current number of transactions and totals by Issuers (by card type). 'TOTAL ' will be used as issuer name and sent in "Company name" field for overall totals (totals for all issuers). If cashier number is sent from the ECR, totals for particular cashier will be sent. Otherwise terminal will return totals for all cashiers.
- In case of Current Issuer Transactions & Totals request the terminal will return all transactions data for particular Issuer (and cashier). Then terminal will return current totals for same Issuer (and cashier). Transaction type in response will be different for transactions and totals. In case of transactions, terminal type will be equal to transaction type obtained during processing of original transaction. For example Transaction type for Sale will be 01, for Cash 03, etc. Transaction type for current totals will match with one from request (34 in this case). This procedure will be repeated for every particular Issuer (card type). Finally, terminal will return overall totals and issuer name 'TOTAL '.If cashier number is sent from the ECR, transactions and totals for particular cashier will be sent. Otherwise terminal will return transactions and totals for all cashiers
- Payservices Data in response message are TLV based. TLV format is similar to EMV data (description of TLV format is above). Tags and fields are described in Appendix A of this document.
- Available balance – available amount on cardhodler account. First position is sign (+ positive balance, - negative balance). Rest of message is formatted amount.
- Loyalty Data in response message are TLV based. TLV format is similar to EMV data (description of TLV format is above). Tags and fields are described in Appendix B of this document.
- Personal/Vehicle Card Data in response message are TLV based. TLV format is similar to EMV data (description of TLV format is above). Tags and fields are described in Appendix C (Personal Card Data) and Appendix D (Vehicle Card Data) of this document.
- Transaction approved – transaction is approved when value of Transaction flag is 01 or 02 as stated in the table.

2.2.10.2. Message 20: Hold (**DEPRECATED** – Extended Hold message 25 should be used)

This message is send back from the POS terminal when it performs a long operation as authorization call or as a ticket printing. The external application will wait for the response during 2 minutes.

	Name	Condition	Size	Format	Value
1.	Identifier	Mandatory	2	Numeric	Always **20**
2.	00	Mandatory	2	Numeric	
3.	00	Mandatory	2	Numeric	
4.	00	Mandatory	2	Numeric	
5.	Display message	Optional	25 max	Alpha-numeric	Message that will be displayed during the hold (ex : Please Wait)

2.2.10.3. Message 22: Error (**DEPRECATED** – Extended Error message 26 should be used)

This message is send back from the POS terminal as consequence of a wrong request message or an application error condition.

	Name	Condition	Size	Format	Value
1.	Identifier	Mandatory	2	Numeric	Always **22**

	Name	Condition	Size	Format	Value
2.	00	Mandatory	2	Numeric	
3.	00	Mandatory	2	Numeric	
4.	00	Mandatory	2	Numeric	
5.	Display message	Optional	25 max	Alpha-numeric	Message that will be displayed to operator screen

2.2.10.4. Message 30: Card data

This message is sent from the POS terminal when transaction is activated by the terminal. The external application will accept card data and starts transaction (optionaly).

	Name	Condition	Size	Format	Default Value	Value
1.	**Identifier**	Mandatory	2	Numeric	10	Always 3**0**
2.	**00**	Mandatory	2	Numeric		
3.	**00**	Mandatory	2	Numeric		
4.	**00**	Mandatory	2	Numeric		
5.	**Card Data source**	Mandatory	1	Alpha-numeric		**M** : Magnetic stripe **S** : Smart Card **K:** Manual PAN entry **C:** Contactless **U**: Unknown
6.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	
7.	**Card Hash**	Optionaly	20	Alpha-numeric		Hashed card number
8.	**Field separator**	Mandatory	1	FS (chr28)	FS (chr28)	

2.2.10.5. Message 23: Cancel Current Transaction

This message is send to the POS terminal to cancel current transaction which can be done only until card is not read

	Name	Condition	Size	Format	Value
1.	Identifier	Mandatory	2	Numeric	Always **23**
2.	00	Mandatory	2	Numeric	
3.	00	Mandatory	2	Numeric	
4.	00	Mandatory	2	Numeric	

2.2.10.6. Message 24: Cancel Current Transaction response

This message is sent from the POS terminal as response to cancel current transaction request. The external application will receive cancel current transaction status.

	Name	Condition	Size	Format	Value
1.	Identifier	Mandatory	2	Numeric	Always **24**
2.	00	Mandatory	2	Numeric	
3.	00	Mandatory	2	Numeric	
4.	00	Mandatory	2	Numeric	
5.	Response message	Optional	25 max	Alpha-numeric	Message that will be sent to external application

2.2.10.7. Message 25: Extended Hold

If extended hold messages are enabled by the terminal configuration, this message is send back from the POS terminal when it performs a long operation as authorization call or as a ticket printing. The external application will wait for the response during 2 minutes.

	Name	Condition	Size	Format	Value
1.	Identifier	Mandatory	2	Numeric	Always **25**
2.	00	Mandatory	2	Numeric	
3.	00	Mandatory	2	Numeric	
4.	00	Mandatory	2	Numeric	
5.	Display message	Mandatory	25 max	Alpha-numeric	Message that will be displayed during the hold (ex : Please Wait)
6.	Field Separator	Mandatory	1	FS (chr 28)	
7.	Code	Mandatory	3 max	Alpha-numeric	Alpha numeric code as given in the table below
8.	Field Separator	Mandatory	1	FS (chr 28)	

Codes used in Extended Hold messages:

ECR_HOLD_PLEASEWAIT	"50"
ECR_HOLD_INSERTCARD	"51"
ECR_HOLD_REMOVECARD	"52"
ECR_HOLD_SWIPECARD	"53"
ECR_HOLD_CARDINSERTED	**"54"**
ECR_HOLD_ENTERPIN	"55"
ECR_HOLD_PINOK	"56"
ECR_HOLD_HOSTCOMM	"57"
ECR_HOLD_INPUTDATA	"58"
ECR_HOLD_CANCELED	"59"
ECR_HOLD_PRESENTCARD	"60"

Code 54 - ECR_HOLD_CARDINSERTED indicate that the card is read and the transaction is started. After that, cancelation of current transaction is not possible anymore.

2.2.10.8. Message 26: Extended Error

If extended error messages are enabled by the terminal configuration, this message is send back from the POS terminal as consequence of a wrong request message or an application error condition.

	Name	Condition	Size	Format	Value
1.	Identifier	Mandatory	2	Numeric	Always **26**
2.	00	Mandatory	2	Numeric	
3.	00	Mandatory	2	Numeric	
4.	00	Mandatory	2	Numeric	
5.	Display message	Mandatory	25 max	Alpha-numeric	Message that will be displayed during the hold (ex : Please Wait)
6.	Field Separator	Mandatory	1	FS (chr 28)	
7.	Code	Mandatory	3 max	Alpha-numeric	Alpha numeric code as given in the table below
8.	Field Separator	Mandatory	1	FS (chr 28)	

Codes used in Extended Error messages:

ECR_ERR_OK	"00"
ECR_ERR_FAILED	"01"
ECR_ERR_SETTLEREQUIRED	"02"
ECR_ERR_TRANSNOTALLOWED	"03"
ECR_ERR_UNSUCCESSFUL	"04"
ECR_ERR_INVALIDCARD	"05"
ECR_ERR_INVALIDENTRY	"06"
ECR_ERR_ERROR	"07"
ECR_ERR_UNSUPPORTEDCARD	"08"
ECR_ERR_CANCELED	"09"
ECR_ERR_NOPAPER	"10"
ECR_ERR_REMOVECARD	"11"
ECR_ERR_CARDEXPIRED	"12"
ECR_ERR_DECLINED	"13"
ECR_ERR_MANPANNOTALLOWED	"14"
ECR_ERR_NOTLOGGEDON	"15"
ECR_ERR_CHIPNOTSUPPORTED	"16"
ECR_ERR_TRANSNOTFOUND	"17"
ECR_ERR_CANCELEDBYECR	"18"
ECR_ERR_INITFAILED	"19"
ECR_ERR_SETTLEMENTFAILED	"20"
ECR_ERR_LOGONFAILED	"21"
ECR_ERR_INITINPROGRESS	"22"

2.2.10.9. Message 27: Send Mifare Command

If Mifare card is approached on transaction request and the terminal responded accordingly, ECR should continue with Mifare commands that should be sent to the card.

	Name	Condition	Size	Format	Value
1.	Identifier	Mandatory	2	Numeric	Always **27**
2.	00	Mandatory	2	Numeric	
3.	00	Mandatory	2	Numeric	
4.	00	Mandatory	2	Numeric	
5.	Command Type	Mandatory	2	Numeric	Mifare command type 00 – Authenticate Sector 01 – Read Block 02 – Write Block 03 – Read Sector 04 – Write Sector 99 – Operation complete
6.	Command Data Length	Mandatory	4	Hex	Length of the data needed for the command (hexadecimal encoded, converted to ASCII)
7.	Command Data	Mandatory	1000 max	Alpha-numeric	Command data

Command data content depends on the command type and it is TLV based. TLV format is similar to EMV data (description of TLV format is above). Tags and fields are described in below.

Format:

- *N – numeric field*
- *AN – alpha-numeric field*
- *Hex – hexadecimal numeric field*

Size:

- *xx – fixed length data element*
- *...xx – variable length data element (less or equal to xx)*

Important: Command data is converted to ASCII before transmission to avoid the appearance of non printable chatacters in serial communication. Size of data in the message is doubled and the command data length in the message indicates the real number of bytes before conversion.

Tag	Name	Size	Format	Description
SC	Sector	1	Hex	Number of the sector of the card (present in all commands)
BL	Block	1	Hex	Number of the block of the card (present in Read Block and Write Block commands)
KT	Key Type	1	Hex	Authentication key type (Key A – 0x60 or Key B – 0x61) (present in Authenticate Sector, Read Sector and Write Sector commands)
KY	Authentication key	...16	Hex	Authentication key (present in Authenticate Sector, Read Sector and Write Sector commands)

WD	Write data	...64	Hex	Data to be written (present in Write Block and Write Sector commands)

2.2.10.10. Message 28: Response to Mifare Command

Card response on Mifare command is transferred by POS terminal to ECR.

	Name	Condition	Size	Format	Value
1.	Identifier	Mandatory	2	Numeric	Always **28**
2.	00	Mandatory	2	Numeric	
3.	00	Mandatory	2	Numeric	
4.	00	Mandatory	2	Numeric	
5.	Command Response Status	Mandatory	2	Numeric	Status of the command execution: 00 – Successful 01 – Error
6.	Response Length	Mandatory	4	Hex	Length of the response from Mifare card
7.	Response Data	Mandatory	1000 max	Alpha-numeric	Response data

In case of Read Block and Read Sector commands are sent and reading is successful, the response data field will contain the value of read data. Otherwise, the response data will be empty and the length provided will be zero.

Important: Response data is converted to ASCII before transmission to avoid the appearance of non printable chatacters in serial communication. Size of data in the message is doubled and the response data length in the message indicates the real number of bytes before conversion.

2.2.10.11. Message 31: Function processing request

This message is send to the POS terminal to start terminal function

	Name	Condition	Size	Format	Value
1.	Identifier	Mandatory	2	Numeric	Always **31**
2.	Function number	Mandatory	3	Numeric	
3.	Field separator	Mandatory	1	FS (chr28)	FS (chr28)
4.	Pasword	Optional	10 max	Numeric	Password needed to process some transactions
5.	Field separator	Mandatory	1	FS (chr28)	FS (chr28)
6.	Additional data	Optional	100 max	Alpha-numeric	Additional information
7.	Field separator	Mandatory	1	FS (chr28)	FS (chr28)

Message example: 020233313039321C1C1C0326

2.2.10.12. Message 32: Function processing response

This message is sent from the POS terminal as response to function processing request. The external application will receive function status and additional data if applicable.

	Name	Condition	Size	Format	Value
1.	Identifier	Mandatory	2	Numeric	Always **32**
2.	Status	Mandatory	1	Numeric	0 – Successful 1 – Error
3.	Field separator	Mandatory	1	FS (chr28)	FS (chr28)
4.	Response message	Optional	1000 max	Alpha-numeric	Message that will be sent to external application
5.	Field separator	Mandatory	1	FS (chr28)	FS (chr28)

Message example:
023332301C5445524D494E414C2049443A203632323130524454 0A4150503A2041535345
434F504F530A4255494C443A203230303232362F30310A53444B2056657273696F6E3A20
392E31382E310A454D562056657273696F6E3A2030352E33380A434C204D434152443A2
0343234312E303737330A434C20564953413A20333335302E303430380A4150502054595
0453A20474F414C2054320A434C20414D45583A20343239372E303630330A434C205A49
503A20333335322E303130310A434C20445041533A203834343535373033313 10A434C20
515041533A204E410A434C20505552453A204E410A434C45535320535550504F52543A20
5945530A4F50454E57415920535550504F52543A204E4F0A545241434520454E41424C45
443A205945530A4859504552205355 0504F52543A205945530A535044482053555 0504F5
2543A205945530A434F525445582053555 0504F52543A205945530A424F52494341205355
50504F52543A205945530A5450494920535550504F52543A205945530A1C0375

2.2.10.13. Message 33: Intermediate transportation transaction information

In case of deferred authorisation is requested for transportation (Radcom), the terminal sends intermediate information about transaction in composite Specific Radcom / transportation related data including hashed card number to inform the validator that the card is tapped.Validator in the bus will check if this card is present on transportation temporary stop list or the card is used as an identification. The response to the terminal is described in the next section.

	Name	Condition	Size	Format	Value
1.	Identifier	Mandatory	2	Numeric	Always **33**
2.	00	Mandatory	2	Numeric	
3.	00	Mandatory	2	Numeric	
4.	00	Mandatory	2	Numeric	
5.	Transaction number	Mandatory	6	Numeric	Transaction number assigned by the terminal
6.	Specific Radcom / transportation related data	Mandatory	1000 max	Aplha-numeric	Specific Radcom / transportation related data
7.	Field separator	Mandatory	1	FS (chr28)	FS (chr28)

2.2.10.14. Message 34: Intermediate transportation transaction information response

In case of deferred authorisation is requested for transportation (Radcom), validator in the bus will check if this card is present on transportation temporary stop list or the card is used as an identification. The response to the terminal is sent to indicatate status of .

	Name	Condition	Size	Format	Value
1.	Identifier	Mandatory	2	Numeric	Always **34**
2.	00	Mandatory	2	Numeric	
3.	00	Mandatory	2	Numeric	
4.	00	Mandatory	2	Numeric	
5.	Status	Mandatory	2	Numeric	00- Continue transaction with deferred authorisation 01- Card is on transportation blacklist – stop transaction 02- Card is used as identification (i.e. monthly ticket) – transaction approved locally
6.	Specific Radcom / transportation related data	Optional	1000 max	Aplha-numeric	Specific Radcom / transportation related data
7.	Field separator	Mandatory	1	FS (chr28)	FS (chr28)

2.2.10.15. Message 29: Transportation deferred authorisation indication

In case of deferred authorisation, when the terminal successfully transmits the message to payment gateway, the terminal informs the validator. After that transportation back offce can require nominal authorisations from payment gateway.
Transaction number is unique per transaction throughout the entire flow.
The terminal does not expect any response on this message.

	Name	Condition	Size	Format	Value
1.	Identifier	Mandatory	2	Numeric	Always **29**
2.	00	Mandatory	2	Numeric	
3.	00	Mandatory	2	Numeric	
4.	00	Mandatory	2	Numeric	
5.	Transaction number	Mandatory	6	Numeric	Transaction number assigned by the terminal
6.	Specific Radcom / transportation related data	Optional	1000 max	Aplha-numeric	Specific Radcom / transportation related data
7.	Field separator	Mandatory	1	FS (chr28)	FS (chr28)

APPENDIX A

PayServices Tags and Fields

Format:

- *N – numeric field*
- *AN – alpha-numeric field*

Size:

- *xx – fixed length data element*
- *...xx – variable length data element (less or equal to xx)*

Tag	Name	Size	Format	Request	Response
PC	Product Code	2	N	Mandatory	Mandatory
AC	Voucher Activation Code	...16	N		Optional
AD	Voucher Activation Date	6	AN		Optional
AT	Voucher Air Time	12	AN		Optional
SN	Voucher Serial Number	...16	AN		Optional
VD	Voucher Validity	...4	AN		Optional
AM	Formatted Total Amount	...17	AN		Optional
DA	Formatted Discount Amount	...17	AN		Optional
ET	Event Title	...80	AN		Optional
I3	Info Area	...400	AN		Optional
F1	Additional Field 1 Value	...20	AN	Optional	
F2	Additional Field 2 Value	...20	AN	Optional	
F3	Additional Field 3 Value	...20	AN	Optional	
F4	Additional Field 4 Value	...20	AN	Optional	

F5	Additional Field 5 Value	...20	AN	Optional	

Field Description:

- Product Code – ID of service used. Product Code must be unique per service.
- Additional Field 1..5 – additional data if necessary for particular service. Service could require addiotonal data to be provided. In that case terminal require additional filed value.
- Voucher Activation Code – activation code for Topup voucher. It should be decripted between provider and ECR.
- Voucher Activation Date – date of Voucher validity. Voucher could not be activated after this date. Activation Date format is YYMMDD.
- Voucher Air Time – voucher amount.
- Voucher Serial Number – unique serial number (ID) of voucher.
- Voucher Validity – days of voucher validity. For information purpose only. Voucher could not be activated after Activation Date.
- Formatted Total Amount – formatted amount with discount if there is discount on this service.
- Formatted Discount Amount – discount amount calclated by provider if discount processing is enabled for this service.
- Event Title – services name returned by service provider.
- Info Area – additional data sent by service provider (additional instructions, advertising messages, etc.)

APPENDIX B

Loyalty Data Tags and Fields

Format:

- *N – numeric field*
- *AN – alpha-numeric field*

Size:

- *xx – fixed length data element*
- *...xx – variable length data element (less or equal to xx)*

Tag	Name	Size	Format	Response
TI	Terminal ID	...16	AN	Mandatory
MI	Merchant ID	...15	AN	Optional
PO	Payment Option	2	N	Mandatory
IS	Issuer Name	...10	AN	Optional
PN	PAN	4	N	Mandatory
BN	Batch Number	6	N	Mandatory
IN	Invoice Number	6	N	Mandatory
AU	Authorization Code	...6	AN	Optional
CI	Customer ID	...20	AN	Optional
OA	Original Amount	...12	N	Optional
AM	Amount	...12	N	Mandatory
I3	Info Area	...400	AN	Optional
LV	Loyalty Voucher Data	...400	AN	Optional

Field Description:

- Terminal ID – Terminal ID of Loyalty Acquirer.
- Merchant ID – Merchant ID of Loyalty Acquirer.
- Payment Option – Payment option of the transaction comparing to loyalty card. Possible values are: 01 – Same card is used for loyalty and payment; 02 – Other card is used for payment; 03 – Bill is paid in cash.
- Issuer name – Issuer/card name assigned to loyalty card/range.
- PAN – Last 4 digits of PAN of Loyalty card.
- Batch Number – Current batch number of loyalty acquirer batch.
- Invoice Number – Unique invoice number of loyalty transaction.
- Authorization Code – Authorization/approval code for loyalty transaction.
- Customer ID – Loyalty customer identification number.
- Original Amount – Original transaction amount, before discount is calculated. It is sent only in case of discount is present..
- Amount – Total amount, including possible discount.
- Info Area – additional data sent by loyalty server (additional instructions, advertising messages, etc.)
- Loyalty Voucher Data – Printable voucher related data including, voucher activation code, validity, description etc. Activation code is always preceded with "A/C:".

APPENDIX C

Personal Card data Tags and Fields

#No	Tag	Description	Type
1	**09**	Country code	text
2	**0A**	Personal card number	text
3	**0B**	Type	text
4	**0C**	Type and number	text
5	**0D**	Date of issue	text
6	**0E**	Date of expiry	text
7	**0F**	Issued by (ISO)	text
8	**10**	Code 1	text
9	**11**	Code 2	text
10	**16**	Personal number (JMBG)	text
11	**17**	Surname (ISO)	text
12	**18**	Name (ISO)	text
13	**19**	Parent name (ISO)	text
14	**1A**	Gender (ISO)	text
15	**1B**	Birth place (ISO)	text
16	**1C**	Birth municipality (ISO)	text
17	**1D**	Birth country (ISO)	text
18	**1E**	Birth date	text

19	**1F00**	Birth country code	text
20	**20**	Residence country code	text
21	**21**	Residence municipality (ISO)	text
22	**22**	Residence place (ISO)	text
23	**23**	Residence street (ISO)	text
24	**24**	Residence number (ISO)	text
25	**25**	Address line 3 (ISO)	text
26	**26**	Address line 4 (ISO)	text
27	**27**	Address line 5 (ISO)	text
28	**28**	Address line 6 (ISO)	text

*All data is encoded by ISO/IEC 8859-2 standard;

**Empty tags will not be present in response;

***Only for Serbian personal card;

APPENDIX D

Vehicle Card data Tags and Fields

#No	Tag	Description	Type
1	**80**	Version	text
2	**9F33**	Country	text
3	**9F34**	Label	text
4	**9F35**	Issued by	text
5	**9F36**	Issued by	text
6	**9F37**	Character set	text
7	**9F38**	Document number	text
8	**81**	Registration number	text
9	**82**	First registration date	text
10	**83**	Registration owner name	text
11	**84**	Registration owner first name	text
12	**85**	Registration owner address	text
13	**86**	Registration owner flag	text
14	**87**	Vehicle mark	text
15	**88**	Vehicle type	text
16	**89**	Vehicle label	text

17	**8A**	Vehicle identification number	text
18	**8B**	Vehicle maximum mass	text
19	**8C**	Vehicle mass	text
20	**8D**	Validity	text
21	**8E**	Registration date	text
22	**8F**	Homologation number	text
23	**90**	Motor volume	text
24	**91**	Motor power	text
25	**92**	Motor fuel	text
26	**93**	Power Mass	text
27	**93**	Seating places	text
28	**94**	Standing places	text
29	**96**	Maximum mass	text
30	**97**	Maximum mass for vehicle system	text
31	**98**	Category	text
32	**99**	Number of axis	text
33	**9A**	Axis distance	text
34	**9F1F00**	Mass axis 1	text
35	**9F20**	Mass axis 2	text
36	**9F21**	Mass axis 3	text
37	**9F22**	Mass axis 4	text
38	**9F23**	Mass axis 5	text
39	**9B**	Trailer mass locked	text
40	**9C**	Trailer mass unlocked	text
41	**9D**	Motor moment	text
42	**9E**	Motor identification number	text
43	**9F24**	Color	text
44	**9F25**	Maximum speed	text
45	**9F26**	Noise standing	text
46	**9F27**	Noise moment	text
47	**9F28**	Noise driving	text
48	**9F29**	Gas emission CO	text
49	**9F2A**	Gas emission HC	text
50	**9F2B**	Gas emission NOx	text
51	**9F2C**	Gas emission HC + NOx	text
52	**9F2D**	Gas emission Particles (diesel)	text
53	**9F2E**	Gas emission Corrected absorbing quotient (diesel)	text
54	**9F2F**	Gas emission CO2	text
55	**9F30**	Combined consumption	text
56	**9F31**	Environment category	text
57	**9F32**	Fuel tank	text

58	**C1**	Forbidden to sell before	text
59	**C2**	Personal number (JMBG)	text
60	**C5**	Production year	text
61	**C9**	Serial number	text

*All data iss encoded by ISO/IEC 8859-2 standard;

**Empty tags will not be present in response;

***Compliant with FIA standard, it may also be possible to use for vehicle cards made outside Serbia.

APPENDIX E

Part of composite Specific Radcom / transportation related data.

#No	Tag	Description
1	**HP**	Hashed pan number – sent by the terminal. Validator uses this to check transportation stop list. Hash alorithm must be strong enough to avoid various attacks.
2	**TE**	Ticket expiry – provided by the validator and might be compared to card expiry on the terminal when seasonal ticket is bought. If this information is present in the request, the terminal must check the card expiry date and decline transaction if the card expires before the ticket validity.
3	**CS**	Card Serial Number / Unique ID (UID) if it might be obtained form the physical card/device
4	**PA**	Payment Account Reference (PAR) – EMV tag 9F24 - It will be provided if the card and kernel return such information.
5	**DT**	Card device type / form factor – EMV tag 9F6E for Visa and Mastercard. It may indicate different card types: real card, mobile phone, key fob, watch etc.
6	**PR**	Passenger ID refference - 10 alpha numeric characters long and should be unique for each card tapped. This information will be sent to host and it shall be seen in message form host to the cardholder or bank statement.
7	**AT**	Authenticity token – sent by the terminal. Unique terminal ID
8	**HM**	Hashed pan number – sent by the terminal. Validator uses this to check card as credentials functionality. Hash alorithm must be strong enough to avoid various attacks.
9		